

Project MAGI: Multi-terrain Autonomous Ground Intelligence

Executive Summary

This document details the complete architectural and developmental history of Project MAGI (formerly MAYA). The project centers on the development of an autonomous, 12-DOF quadruped robotic platform designed for precision agriculture in Indian farming environments. The platform's functional scope shifted during development from a pesticide sprayer to a precision disease classification and mapping system, focusing on crops like peas and cotton in regions like Maharashtra.

I. Evolution and Core Concept

The project, initially named MAYA (Multi-terrain Autonomous Yielding Agent), was officially rebranded as MAGI (Multi-terrain Autonomous Ground Intelligence). The initial concept envisioned an autonomous quadruped robot for hyper-precision pest control via targeted micro-dosing. However, the functional scope was deliberately shifted to focus on precision disease classification and crop health mapping.

The Quadruped Advantage

The 12-DOF quadrupedal design is a fundamental choice to overcome the limitations of traditional wheeled or tracked robots in unstructured agricultural environments. Legged locomotion provides superior mobility over deformable terrain, uneven ground, and dense vegetation. Crucially, the quadruped form factor enables under-canopy operations, granting access to the lower stems and undersides of leaves where pests and fungal diseases often proliferate—areas inaccessible to top-down spraying methods or aerial drones.

II. Technical Architecture

Hardware Specifications

Component	Specification	Function
Primary Controller	Raspberry Pi 4B	High-level processing, edge-AI vision, and system orchestration. Runs Python 3.13.
Locomotion Controller	ESP32	Low-level locomotion control and Inverse Kinematics for the 12-DOF chassis.
Actuators	12x MG996R Servos	High-torque servos driving the quadruped's joints.
Power Source	Custom Battery Pack (3S 5200mAh 25C LiPo / 12,000 mAh 2S4P)	Provides power to the system.
Power Regulation	15A Buck Converters	Regulates voltage to 6V for servos and 5V/3.3V for logic controllers.
Vision Sensor	NoIR Camera	Captures images for multi-spectral analysis.

Power Distribution & Management

Power management is a critical aspect of the MAGI platform. The system utilizes a custom battery solution, initially noted as a 12,000 mAh 2S4P pack, and later specified as a 3S 5200mAh 25C LiPo drone battery. A crucial design correction involved the power architecture for the MG996R servos: the voltage requirement was shifted from 10-12V down to 6V via 15A buck converters to prevent hardware damage and mitigate smoke risks.

III. Software and AI Infrastructure

Edge-AI Vision Pipeline

The perception capabilities of MAGI are driven by a multi-spectral vision pipeline designed for real-time pathology detection. The system utilizes a quantized MobileNetV2 model for efficient edge-AI processing on the Raspberry Pi. This pipeline calculates NDVI (Normalized Difference Vegetation Index) indices, enabling early stress detection and disease classification in crops. The vision processing is implemented using a GStreamer-based pipeline.

Control and Navigation

The robot's navigation and orchestration are managed via a ROS 2 software stack. Specialized kinematics algorithms have been implemented to ensure stability on uneven, loamy agricultural terrain. The control architecture strategically segregates high-level, non-deterministic tasks (managed by the Raspberry Pi) from low-level, time-critical operations like servo control (managed by the ESP32). A Semantic SLAM navigation system was conceptualized to build context-aware maps using crop rows as landmarks.

IV. Development Methodologies

The development of MAGI has involved advanced methodologies and simulated environments:

- Extensive use of containerized development environments, including Docker-based multi-container architectures.
- Integration of VS Code with the Raspberry Pi for remote development.
- Development of a multi-spectral vision pipeline in Google Colab to simulate real-time NDVI computation and disease classification, utilizing cross-spectral image registration scripts.
- Testing of hardware-level pulse-width modulation (PWM) and RC circuit-based transitions on the Raspberry Pi for visual effects.

V. Future Directions and Advanced Concepts

Exploration into advanced locomotive concepts is ongoing. This includes research into wheeled-legged hybrid systems and gearless electromagnetic motor designs for robotic appendages. Conceptualization of "skating" kinematics and the use of omni-wheel systems that act as independent motors are being evaluated to further enhance the platform's maneuverability and efficiency.